

Mercedes-Benz Attention Assist

Attention Assist development was a four-year test marathon with about 1,000,000 km driven by more than 700 test drivers. The acceptance of the system amongst test drivers and customers was predominantly good. In the tests, Mercedes scientists found that drowsy drivers had trouble steering a precise course in their lane, thus making minor steering errors. This effect occurred at an early stage when drowsiness kicked in—often before a dangerous situation in which the driver fell asleep momentarily.

Electroencephalogram (EEG) for brain wave monitoring

Brain waves are formed within brain cells to communicate with each other. These reflect the activity of the brain by emitting electrochemical impulses of various frequencies. Brain waves are categorised as follows:

Beta—When we are consciously alert, tensed, agitated or afraid, the emitted frequency is 13 to 60 pulses per second.

Alpha—When we are mentally relaxed, although aware of what is happening, the emitted frequency is 7-13 Hz.

Theta—When the alertness is reduced, i.e., in meditation, the emitted frequency is 4-7 pulses per second.

Delta—It is the state of unconsciousness, deep sleep or catalepsy, emitting 0.1-4Hz pulses.

Electroencephalogram is a high-end medical electronic equipment that is used to monitor the electrical activity of brain by firing neurons.

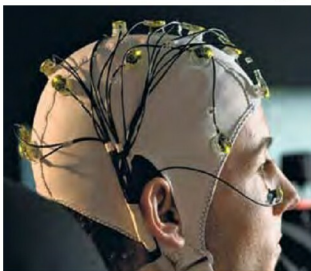
drowsiness and fatigue. One of the most common methods is monitoring of the blinking of eyelids. Another method is monitoring of the lane-keeping performance.

After several years of research, Mercedes engineers found that drowsy drivers make more minor steering errors, which they often correct very rapidly and abruptly in characteristic ways. In the drowsiness detection method used by them, the quick and abrupt corrections are recognised by a highly sensitive steering-angle sensor.

To identify the point where the driver actually slips into weariness from an alertness, the engineers at Mercedes fitted many test drivers with brainwave-monitoring skullcaps to obtain their electroencephalograms (EEG).

Data acquisition and processing

The quality of drowsiness detection is highly dependent on how accurate the parameters are acquired and analysed. Attention Assist, which comes as part of the standard equipment in the E-class, uses several highly sensitive sensors. These sensors continu-



Attention Assist test sensors

ously monitor more than 70 different parameters. External influences such as wind and road irregularities are also taken into consideration. In addition, vehicle speed, lateral acceleration and longitudinal acceleration are continuously monitored.

How does Attention Assist work?

Based on the huge amount of data acquired by various sensors, Attention Assist identifies the driver's driving profile in the first few minutes of every

trip. This profile is then compared, by the car's electronic control unit, with the current sensor data and the prevailing driving situation. This continuous monitoring is crucial for detecting floating transition from awareness to drowsiness and for giving timely warning to the driver.

When the warning threshold is reached, or a suspicious behaviour is detected, the system triggers an acoustic warning signal and a text message (coffee mug symbol with "Attention Assist Pause!") in the instrument cluster.

Working conditions

The system is active at a speed of 80-180 kmph. It creates the driver's profile through analysis and classification of his steering and driving behaviour in order to determine the individual warning threshold. A complete profile is typically formed in 20-30 minutes

depending on the road and traffic conditions.

Before formation of the complete driver profile, a standard threshold is used. Unfortunately, you cannot save a driver's profile because it may vary.

The system works positively on highways and expressways in Western countries. For India, the average speed on highways in different states varies from 65 kmph to 90 kmph. Only some expressways have a speed limit of 100-

120 kmph. So the system is inactive for most of the drive time in India. "Technically, it is possible to customise the system but the feasibility has not been determined yet," shares Dr Breuer.

With the present conditions of roads, speed limits and traffic, we are left to think "how can Attention Assist work in India if it is active only at 80-180 kmph?" ●

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